

STA355F: Theory of Statistical Practice

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1 Additional probability review

1.1 Probabilistic convergence and arithmetic operations

Theorem 2.1. Suppose that $(X_n)_{n \geq 1}$ and $(Y_n)_{n \geq 1}$ are sequences of random variables, and X, Y are random variables.

1. If $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} Y$, then $X_n + Y_n \xrightarrow{p} X + Y$ and $X_n Y_n \xrightarrow{p} XY$.
2. If $X_n \xrightarrow{\mathbb{L}^p} X$ and $Y_n \xrightarrow{\mathbb{L}^p} Y$, then $X_n + Y_n \xrightarrow{\mathbb{L}^p} X + Y$, but this does not necessarily hold for the product $X_n Y_n$.
3. (Slutsky's theorem) If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$ for some constant c , then $X_n + Y_n \xrightarrow{d} X + c$, $X_n Y_n \xrightarrow{d} cX$, and $X_n/Y_n \xrightarrow{d} X/c$ if $c \neq 0$.

Note: Slutsky's theorem does not hold true if Y_n converges in distribution to a non-degenerate random variable Y instead of a constant. On the other hand, it does not require any dependence structure between X_n and Y_n .

Intuition about Slutsky: $X_n \xrightarrow{d} X$ only tells you about the marginal distribution of X_n . It does not tell you anything about the joint distribution of (X_n, Y_n) . But if $Y_n \xrightarrow{p} c$, you are in a degenerate case, and the joint distribution of (X_n, Y_n) does not matter.

1.2 $o_p()$ and $O_p()$ notation

Notations about limit of sequences: let $(a_n)_{n \geq 1}$ and $(b_n)_{n \geq 1}$ be sequences of real numbers. $b_n > 0$ for all n .

- We say that $a_n = o(b_n)$ if $\lim_{n \rightarrow \infty} a_n/b_n = 0$.
- We say that $a_n = O(b_n)$ if there exists a constant $C > 0$ such that $|a_n| \leq Cb_n$ for all sufficiently large n .

e.g. $a_n = o(1)$ means $a_n \rightarrow 0$, and $a_n = O(1)$ means a_n is bounded.

e.g. when $a_n = 2n^2 + n + 1$ and $b_n = n^2$, we have $a_n = O(b_n)$.

e.g. when $a_n = 1/n$ and $b_n = \log n/n$, we have $a_n = o(b_n)$.

Probabilistic version: Let $(X_n, Y_n)_{n \geq 1}$ be sequences of random variables, and $Y_n > 0$ with probability 1 for all n .

- We say that $X_n = o_p(Y_n)$ if $X_n/Y_n \xrightarrow{p} 0$.
- We say that $X_n = O_p(Y_n)$ if for any $\varepsilon > 0$, there exists a constant $C_\varepsilon > 0$ such that $\mathbb{P}(|X_n| \leq C_\varepsilon Y_n) \geq 1 - \varepsilon$ for all n . (X_n/Y_n is asymptotically tight).

Remarks:

$o_p(1)$ means converges in probability to 0, and $O_p(1)$ means bounded in probability (probabilistic mass is not escaping to infinity as $n \rightarrow \infty$).

Facts:

- $O_P(1) + O_P(1) = O_P(1)$.
- $o_P(1) + o_P(1) = o_P(1)$.
- $O_P(1) \times o_P(1) = o_P(1)$.
- $O_P(1) \times O_P(1) = O_P(1)$.

1.3 Delta method

Motivation: the estimators in practice are not always sample average, but some nonlinear functions related to sample average. We want to understand the asymptotic distribution of these estimators.

Theorem 2.2 (Delta method). *Suppose that $(X_n)_{n \geq 1}$ is a sequence of random variables such that $\sqrt{n}(X_n - \mu) \xrightarrow{d} N(0, \sigma^2)$ for some constant μ and $\sigma^2 > 0$. (X_n themselves do not need to be sample averages.) Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a continuously differentiable function at μ with $g'(\mu) \neq 0$. Then we have*

$$\sqrt{n}(g(X_n) - g(\mu)) \xrightarrow{d} N(0, \sigma^2(g'(\mu))^2).$$

Proof idea: we use Taylor expansion of g at μ :

$$g(X_n) = g(\mu) + g'(\mu)(X_n - \mu) + o_p(|X_n - \mu|).$$

By assumption, we have

$$\sqrt{n}g'(\mu)(X_n - \mu) \xrightarrow{d} N(0, \sigma^2(g'(\mu))^2).$$

So we just need to deal with the residual. Since $\sqrt{n}(X_n - \mu) = O_p(1)$, we have $o_p(|X_n - \mu|) = o_p(1/\sqrt{n})$. Therefore, $\sqrt{n}o_p(|X_n - \mu|) = o_p(1) \xrightarrow{p} 0$. By Slutsky's theorem, we conclude that

$$\sqrt{n}(g(X_n) - g(\mu)) = \sqrt{n}g'(\mu)(X_n - \mu) + \sqrt{n}o_p(|X_n - \mu|) \xrightarrow{d} N(0, \sigma^2(g'(\mu))^2).$$

Note: each steps in above proof can be made more rigorous by expanding the definitions of o_p and O_p .

Remarks:

- Continuous differentiability is necessary. e.g., when $g(x) = |x|$, and $\mu = 0$. Then we have

$$\sqrt{n}(g(X_n) - g(\mu)) = \sqrt{n}|X_n| = |\sqrt{n}X_n| \xrightarrow{d} |Y| \text{ where } Y \sim N(0, \sigma^2).$$

- If $g'(\mu) = 0$, then the middle term in the Taylor expansion vanishes, and we have $g(X_n) - g(\mu) = o_p(|X_n - \mu|) = o_p(1/\sqrt{n})$. Technically, the Delta method result is still valid, as $\sqrt{n}(g(X_n) - g(\mu)) = o_p(1) \xrightarrow{d} 0$, which is a degenerate normal distribution. But this is not useful for characterizing the shape of the distribution. In this case, we need to use higher-order Taylor expansion to get a non-degenerate limiting distribution.

Theorem 2.3 (second-order Delta method). *Suppose that $(X_n)_{n \geq 1}$ is a sequence of random variables such that $\sqrt{n}(X_n - \mu) \xrightarrow{d} N(0, \sigma^2)$ for some constant μ and $\sigma^2 > 0$. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a twice continuously differentiable function at μ with $g'(\mu) = 0$ and $g''(\mu) \neq 0$. Then we have*

$$\frac{2n(g(X_n) - g(\mu))}{g''(\mu)\sigma^2} \xrightarrow{d} \chi_1^2.$$

Here χ_1^2 is the probability distribution of Y^2 where $Y \sim N(0, 1)$.

Proof is similar: use second-order Taylor expansion of g at μ :

$$g(X_n) = g(\mu) + \frac{1}{2}g''(\mu)(X_n - \mu)^2 + o_p(|X_n - \mu|^2).$$

We just use the fact

$$n(X_n - \mu)^2 = (\sqrt{n}(X_n - \mu))^2 \xrightarrow{d} \sigma^2 Y^2 \text{ where } Y \sim N(0, 1).$$

In principle, we can extend this to higher-order Delta method. But in practice, we usually only need first-order or second-order Delta method.

Multivariate extension: Suppose that $(X_n)_{n \geq 1}$ is a sequence of random vectors in $\mathbb{R}^d$ such that $\sqrt{n}(X_n - \mu) \xrightarrow{d} N(0, \Sigma)$ for some constant vector $\mu \in \mathbb{R}^d$ and positive definite covariance matrix Σ . Let $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be a continuously differentiable function at μ with gradient $\nabla g(\mu) \neq 0$. Then we have

$$\sqrt{n}(g(X_n) - g(\mu)) \xrightarrow{d} N(0, \nabla g(\mu)^\top \Sigma \nabla g(\mu)).$$

Remark: we can also extend this to vector-valued functions $g : \mathbb{R}^d \rightarrow \mathbb{R}^k$ with Jacobian matrix $\nabla g(\mu) \in \mathbb{R}^{k \times d}$, and the limiting distribution is multivariate normal with covariance matrix $\nabla g(\mu) \Sigma \nabla g(\mu)^\top \in \mathbb{R}^{k \times k}$.

Remark: we can also extend the second-order Delta method to multivariate case, using Taylor expansion with Hessian matrix $\nabla^2 g(\mu) \in \mathbb{R}^{d \times d}$. In such case, the limiting random variable is a quadratic form of a multivariate normal random vector, which is a generalized version of chi-squared distribution.

2 Statistical models and statistical inference

A “statistical model” is a class of probability distributions.

- In probability theory, we work with a fixed and known probability distributions, and study its properties.
- In statistics, we work with a class of probability distributions, and we do not know which distribution in the class is the true one. We want to learn about the true distribution from data.

Roughly speaking, statistical models can be divided into two categories: parametric models and nonparametric models.

- Parametric models: the class of distributions is indexed by a finite-dimensional parameter $\theta \in \Theta \subseteq \mathbb{R}^d$. For example, the class of normal distributions $\{N(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma^2 > 0\}$ is a parametric model with parameter $\theta = (\mu, \sigma^2) \in \mathbb{R} \times (0, \infty)$.
- A nonparametric model is also “parametrized”, but the parameter is infinite-dimensional. For example, the class of all distributions on $\mathbb{R}$ is a nonparametric model, where the parameter is the cumulative distribution function $F : \mathbb{R} \rightarrow [0, 1]$ that is non-decreasing and right-continuous, which is an infinite-dimensional object.

More examples of nonparametric models:

- The set of all probability density functions p on $[0, 1]$ such that $|p'(x)| \leq 1$ for all $x \in [0, 1]$. (we will revisit this at the end of this semester).
- The set of all probability distribution $\mathbb{P}$ on $\mathbb{R}$ such that $\mathbb{E}_{\mathbb{P}}[|X|^2] < +\infty$.

2.1 Tasks in statistical inference

Overall, the goal of statistics is to learn about “which distribution in the class is the true one” from data. There are three main tasks in statistical inference:

1. Point estimation.
2. Confidence intervals.
3. Hypothesis testing.

2.1.1 Point estimation

In general, given a statistical model $\{\mathbb{P}_{\theta} : \theta \in \Theta\}$, and a functional $T : \Theta \rightarrow \mathbb{R}^k$ that we want to estimate, we want to construct an estimator $\hat{T}$ based on samples such that $\|\hat{T} - T(\theta^*)\|_2$ is small, where θ^* is the true parameter.

Criteria for evaluating estimators: mean-squared error.

$$\text{MSE}_{\theta}(\hat{T}) = \mathbb{E}_{\theta} \left[\left\| \hat{T} - T(\theta) \right\|_2^2 \right].$$

Here the notation $\|x\|_2$ means the Euclidean norm of a vector $x \in \mathbb{R}^k$.

In general, depending on the aspects of estimators that we want to evaluate, we may also use other norms and other loss functions.

Bias-variance decomposition For MSE, we have

$$\text{MSE}_{\theta}(\hat{T}) = \mathbb{E}_{\theta} \left[\left\| \hat{T} - T(\theta) \right\|_2^2 \right] = \underbrace{\left\| \mathbb{E}_{\theta}[\hat{T}] - T(\theta) \right\|_2^2}_{\text{bias}} + \underbrace{\mathbb{E}_{\theta} \left[\left\| \hat{T} - \mathbb{E}_{\theta}[\hat{T}] \right\|_2^2 \right]}_{\text{variance}}.$$

We say that an estimator $\hat{T}$ is unbiased if $\mathbb{E}_{\theta}[\hat{T}] = T(\theta)$ for all $\theta \in \Theta$. In this case, the MSE is the variance of the estimator.

In general, we want to make the optimal trade-off between bias and variance to minimize MSE. This is also related to model selection in machine learning.

e.g. $X_1, X_2, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathbb{P}$, and $\mathbb{E}_{\mathbb{P}}[X^2] < +\infty$. Suppose that we want to estimate the parameters

$$\mu(\mathbb{P}) = \mathbb{E}_{\mathbb{P}}[X], \quad \sigma^2(\mathbb{P}) = \text{var}_{\mathbb{P}}(X).$$

(In the notation above, $\mathbb{P}$ corresponds to θ , and $(\mu(\mathbb{P}), \sigma^2(\mathbb{P}))$ corresponds to $T(\theta)$.)

Canonical choices:

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad \hat{\sigma}_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \hat{\mu}_n)^2.$$

Straightforward calculations show that $\hat{\mu}_n$ and $\hat{\sigma}_n^2$ are unbiased estimators of $\mu(\mathbb{P})$ and $\sigma^2(\mathbb{P})$, respectively.

Remark: we can also normalize $\hat{\sigma}_n^2$ by n instead of $n-1$. In this case, the estimator is biased, but it may have smaller variance. In general, we want to make the optimal trade-off between bias and variance to minimize MSE. Asymptotically, normalizing by n or $n-1$ does not matter, as the difference is negligible when n is large.

Important properties of estimators:

Definition 2.1. Given $X_1, X_2, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathbb{P}$, an estimator $\hat{T}_n$ is said to be consistent if $\hat{T}_n \xrightarrow{p} T(\theta^*)$ as $n \rightarrow \infty$, where θ^* is the true parameter.

Definition 2.2. An estimator $\hat{T}_n$ is said to be asymptotically normal if there exists an increasing sequence of positive numbers $(a_n)_{n \geq 1}$ such that $a_n \rightarrow \infty$ as $n \rightarrow \infty$, and

$$a_n(\hat{T}_n - T(\theta^*)) \xrightarrow{d} N(0, \Sigma),$$

where Σ is a positive definite covariance matrix.

Consistency is a basic requirement, and asymptotic normality is a stronger property that allows us to quantify uncertainty of the estimator.

2.1.2 Confidence sets

Definition 2.3. For fixed $\alpha \in (0, 1)$, a $(1 - \alpha)$ -confidence set for $T(\theta)$ is a random set C_n such that

$$\mathbb{P}_{\theta}(C_n \ni T(\theta)) \geq 1 - \alpha, \quad \forall \theta \in \Theta.$$

Note: θ here is NOT a random variable. The probability is taken with respect to the randomness of the data, which is used to construct the confidence set C_n . A random subset of $\mathbb{R}^k$ that contains the deterministic quantity $T(\theta)$ with probability at least $1 - \alpha$, for every possible $\theta \in \Theta$.

Definition 2.4. Given $X_1, X_2, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathbb{P}$, a $(1 - \alpha)$ asymptotic confidence set for $T(\theta)$ is a random set C_n such that

$$\liminf_{n \rightarrow \infty} \mathbb{P}_{\theta}(C_n \ni T(\theta)) \geq 1 - \alpha, \quad \forall \theta \in \Theta.$$

Example: let $X_1, X_2, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Ber}(\theta)$, for unknown $\theta \in (0, 1)$.

Different choices of (asymptotic) confidence intervals for θ :

- CI via asymptotic normality. From the corollary of CLT we have learned in the previous lecture, we have

$$\frac{\sqrt{n}(\bar{X}_n - \theta)}{\widehat{se}_n} \xrightarrow{d} \mathcal{N}(0, 1),$$

where $\widehat{se}_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$ is the sample standard error.

We can then construct an asymptotic $(1 - \alpha)$ confidence interval for θ as

$$C_n = \left[\bar{X}_n - z_{1-\alpha/2} \cdot \widehat{se}_n / \sqrt{n}, \bar{X}_n + z_{1-\alpha/2} \cdot \widehat{se}_n / \sqrt{n} \right],$$

where $z_{1-\alpha/2}$ is the $(1 - \alpha/2)$ -quantile of the standard normal distribution. This is asymptotically valid (and asymptotically exact):

$$\lim_{n \rightarrow \infty} \mathbb{P}_\theta(C_n \ni \theta) = 1 - \alpha, \quad \forall \theta \in (0, 1).$$

On the other hand, for finite n , this confidence interval may not have the desired coverage probability $1 - \alpha$.

- CI via probability deviation inequalities. e.g., we can use Chebyshev's inequality

$$\mathbb{E}_\theta[(\bar{X}_n - \theta)^2] = \frac{\theta(1 - \theta)}{n} \leq \frac{1}{4n}.$$

By Chebyshev's inequality, we have

$$\mathbb{P}_\theta(|\bar{X}_n - \theta| \geq \varepsilon) \leq \frac{1}{4n\varepsilon^2}.$$

For this to be less than α , we can choose $\varepsilon = \sqrt{1/(4n\alpha)}$. This gives a finite-sample valid confidence interval for θ :

$$C_n = \left[\bar{X}_n - \sqrt{\frac{1}{4n\alpha}}, \bar{X}_n + \sqrt{\frac{1}{4n\alpha}} \right].$$

This is valid for any finite n , but it is conservative, and the length of the confidence interval is typically larger than the asymptotic one.

Using more advanced probability deviation inequalities, we can sometimes achieve “best-of-both-worlds” confidence intervals that are valid for finite n and also have good asymptotic properties.